

Manush Prajwal

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Professional Summary

5X Hackathon Winner (Google, BNMIT, RNSIT, TOCE) & Backend Engineer with a **growth mindset** and passion for building real-world solutions. Experienced in designing scalable distributed systems and microservice architectures using Node.js, NestJS, Kafka, Docker, and PostgreSQL. Author of **11 research publications (8 IEEE conference papers and 3 journal papers)** and recognized as **3rd Runner-Up at the Google GenAI Hackathon 2025**. Strong focus on translating research concepts into scalable engineering systems supporting thousands of users. Eager to collaborate across **Software Engineering & Product roles**

Skills

Backend & Distributed Systems: Node.js, NestJS, REST APIs, Microservices, Kafka, API Gateway, Event-Driven Architecture, Authentication & Authorization

Databases & Infrastructure: PostgreSQL, Supabase, Docker, CI/CD

Languages: Python, JavaScript, TypeScript, SQL, C, C++

Frontend (Secondary): React.js, Next.js

Achievements

- **5X Hackathon Winner** — Google, BNMIT, RNSIT, TOCE
- **3rd Runner-Up — Google GenAI Hackathon 2025** Developed an AI-powered system leveraging generative AI technologies in a national-level hackathon competition.
- **Research Publications** Author of **11 research papers (8 IEEE conference papers and 3 journal papers)** across machine learning systems, biosensor technology, and secure computing architectures.
- **IEEE Xplore Author Profile** <https://ieeexplore.ieee.org/author/576786069332685>

Work Experience

Freelance Backend Developer

tourindia.com

Travel Agency Platform (TourIndia24)

May 2023 – Present

- Owned backend architecture for a production travel platform, designing REST APIs and database schemas for bookings, users, and package workflows.
- Built scalable database-backed services handling concurrent booking traffic while ensuring data consistency.
- Optimized backend performance and request handling, contributing to a 3× traffic increase and 45% growth in completed bookings.

Projects

Cement Mind – Cement Plant Logistics Automation (Hackathon-Winning Project)

cementmindai.com

- Architected distributed, event-driven backend services for ingesting high-frequency IoT sensor data using **Apache Kafka**.
- Implemented Kafka producers and consumer services to decouple ingestion, validation, aggregation, and processing pipelines.
- Built backend API layers integrating AI-based quality detection models with logistics and monitoring systems.
- Designed asynchronous, fault-tolerant data pipelines enabling near real-time industrial decision-making.

Student Hub – Microservice-Based Academic Platform

studenthub.com

- Designed domain-driven microservices using **NestJS**, with clear service boundaries for jobs, issues, announcements, resources, and study groups.
- Built a centralized **API Gateway** handling authentication, authorization, request routing, and rate limiting.
- Implemented stateless backend services enabling horizontal scaling and fault isolation.
- Developed REST APIs and database schemas supporting 1,300+ daily active users with role-based access control.
- Improved scalability and maintainability while reducing resource discovery time by over 85%.

Education

The Oxford College of Engineering

Sept 2023 – Present

B.E. in Computer Science and Engineering

* Completed Introduction to Computer Science course (or equivalent) — Currently in 2nd year of 4-year undergraduate program

Alvas PU College, Moodubidri

2021 – 2023

Pre-University Course in PCMC (Science Stream)